

Introduction:

This Quick Start Guide is designed to help get you started doing development with the ES (Engineering Sample) release of DK-E1C or DK-B1 DevKits. You should be familiar with the development environments for the Alif Ensemble MCU family and how to load apps onto the B1 or E1C MCU device using the Alif Security Toolkit (SETOOLS). Rev C boards have Revision A5 silicon on them.

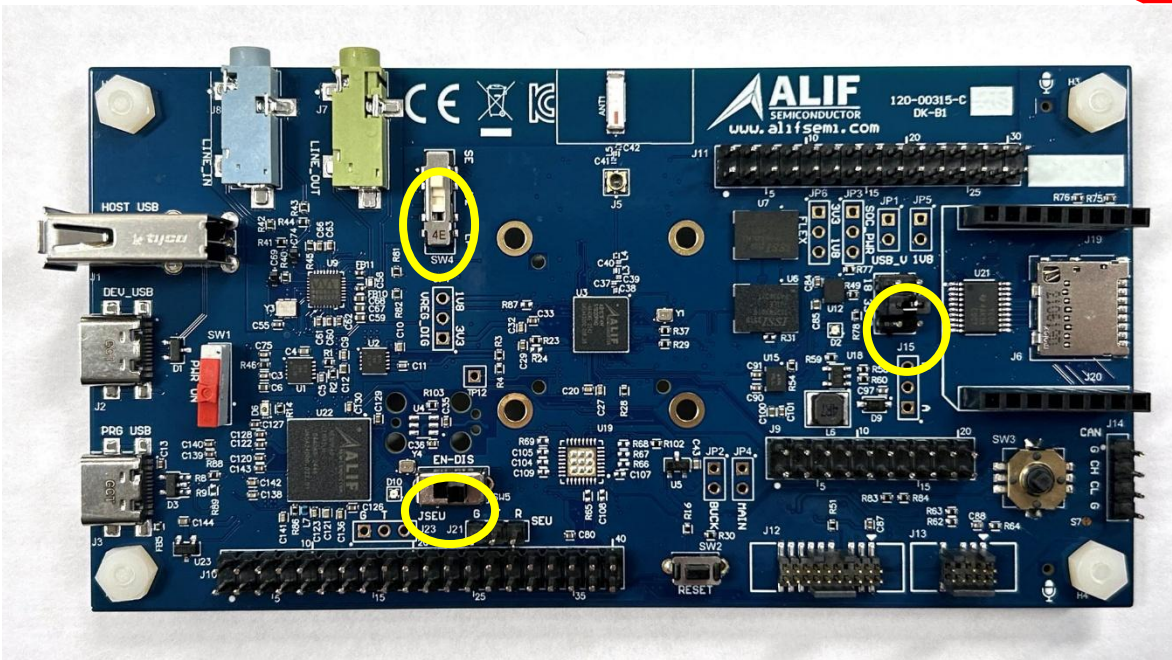
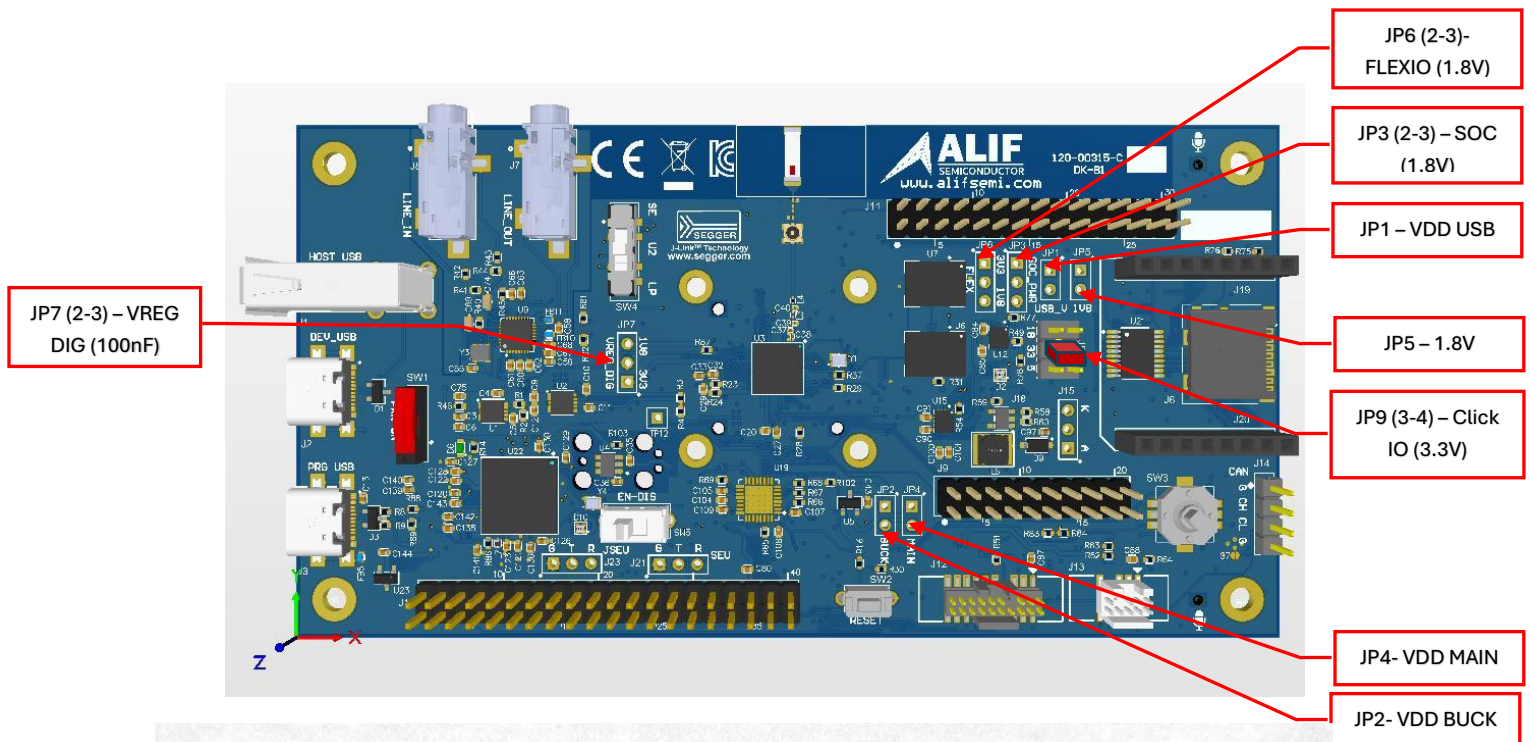
It will go through hardware setup, software that must be installed, and basic development steps of building and debugging a program.

Hardware Setup:

Power to the board can be sourced through either of the connectors PRG_USB or DEV_USB, but for initial development, you should use the PRG_USB port so you have access to the Secure Enclave using the SETOOLS or a user UART.

Verify that the option jumper for JP9 is set in the default position, as shown below, as well as the switch settings for SW4 (UART select) and SW5 (JLOB enable/disable).

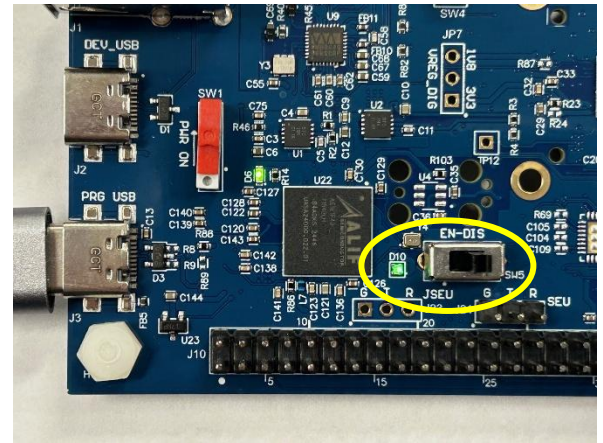
Note: As the kit is delivered, a jumper is installed at jumper post position JP9 and you can use the position as required. However, if any jumper post positions are required other than JP9, you must first cut the shunting trace on the PCB with a small blade that connects the jumper posts together and then solder posts onto the PCB. After this action, you can install and remove a jumper across the posts as required.



Powering Up Your DevKit

Connect a USB-C cable to the PRG_USB connector in the lower left corner and move the red power switch SW1 to the ON position (up towards the “SW1” silkscreen). You should then see a green LED next to U22 in the lower left corner, which indicates that the J-Link On-Board device (U22) is connected to the target E1C or B1 device (U3).

NOTE: Switch SW5 (just to the right of U22) should always be in the right-most position to enable the J-Link On-Board device. Some boards have the silkscreen reversed, as shown in this picture. The left-most position will hold the J-Link On-Board device in reset.



Determining UART Port Assignments

The Ensemble E1C or Balletto B1 DevKit has a USB UART adapter on the board. By default, it is connected to the SEUART that is used by the SETOOLS applications to query device status and program images into the MRAM. It can also be connected to either the UART2 or the LPUART user serial port on the E1C or B1 device through the SW4 UART selection switch.

The optional UART port assignments are discussed in a later section.

SETOOLS Port Discovery

The first time you execute one of the SETOOLS functions, you will be prompted for the required serial port. When the ports are presented, just enter the port name and press [ENTER].

This port data is saved in a local configuration file (isp_config_data.cfg). The next time a command is invoked and this configuration file is present, it will use the parameters from this file.

To override this option simply use the -d option:

```
c:\app-release-exec>maintenance.exe -d
Discover
COM ports detected = 1
-> COM11
Enter port name:COM11
[INFO] COM11 open Serial port success
[INFO] baud rate 55000

Available options:

1 - Device Control
2 - Device Information
3 - MRAM
4 - Utilities
5 - Setting capabilities
6 - ROM

Select an option (Enter to exit):
```

This will force a rediscovery of the Serial ports. If there are no other serial port devices plugged into your system, the COM port shown will be the DevKit port. If there are other COM ports shown, select the one that appears when you plug in the DevKit USB power cable.

UART Errors

There is only one SE-UART on the device. When running the SETOOLS, please ensure you have no other Tera term or putty sessions using the same SE-UART. The following shows the output if the SE-UART is already being used by another program:

```
c:\app-release-exec-65>maintenance -d
COM ports detected = 2
-> COM6
-> COM8
Enter port name:COM6
[ERROR] openSerial could not open port 'COM6': PermissionError(13, 'Access is denied.', None, 5)
[ERROR] isp openSerial failed for COM6
c:\app-release-exec-65>
```


Software:

You will need to download specific versions of the Alif Security Toolkit (SETOOLS), CMSIS Device Family Pack (DFP), and the VS Code template for your Balletto B1 or Ensemble E1C DevKit.

1. SETOOLS v1.109.000 (only for users of revision C kits that have revision A5 silicon).

STOP: Users of kit revision A or B that have revision A3 silicon should NOT update to SETOOLS version 1.109.00 ! Please refer to Documentation Archive Library for previous SETOOLS versions.

The kit revision is identified in the kit part number silkscreen in the upper right corner of the board, as shown in the photo to the right. Kit revision C boards use revision A5 silicon.



Go to the Alif website and, after logging in, click on the head icon at the top right to access your User Account Customer Files area. Select Balletto B1 and look for “Alif Security Toolkit – Executables”. Download the appropriate SETOOLS archive for your platform to a directory on your local machine. Point the tools to this folder for your development by editing the VS Code settings.json file entry for “alif.setools.root” as noted in section 5 below.

Open a terminal window in this SETOOLS directory and run “updateSystemPackage.exe -d”. Select the lowest numbered COM port.

When the tools are installed, they default to an Ensemble E8 device like the one used on the DK-E8 Engineering Sample Alpha Ensemble E8 DevKit. When the tool detects the Balletto B1 or E1C device on your board, it will ask if you want to set them as the default. Reply with “y”. Once you do this, your SETOOLS default is now set to the proper target device for the DK-E1C or DK-B1 DevKits.

```
c:\app-release-exec-102>updateSystemPackage
Burning: System Package in MRAM
Selected Device:
Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B4

Connecting to the target device...
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[INFO] COM6 open Serial port success
Bootloader stage: SERAM
[INFO] Detected Device:
Part# AB1C1F4M51820PH - Rev: A0
- MRAM Base Address: 0x801cd000
Connected target is not the default Part#
Connected target is not the default Revision
Do you want to set this part as default? (y/n): y
Maintenance Mode: Enabled
```

2. Install Visual Studio Code

- Begin by downloading [Visual Studio Code](#)
- While you can install Visual Studio Code using the installation executable, we recommend using the .zip package.
- Using the .zip package will prevent VS Code from keeping its settings on a per-user or per-system basis, allowing multiple installations to coexist without causing clashes in configuration and extensions.
- Unpack the VS Code .zip file to the installation directory of your choice.
- **TIP:** Inside the VS Code directory (where Code.exe is present) create a folder named “data”. If present, VS Code will use this folder to store all the configuration and extensions data, keeping your VS Code installation fully self-contained and portable. This will allow you to have multiple installations present on your system.

3. The VS Code template from GitHub will work as published. It has all board lib contents needed for the E1C and B1 DevKits.

In the top-level directory, (https://github.com/alifsemi/alif_vscode-template), there is a README file with a Quick Start and a link to an online User Guide.

From a command prompt window or the VS Code terminal window, run:

```
git clone https://github.com/alifsemi/alif\_vscode-template <folder_name>
cd <folder_name>
git submodule update --init
```

NOTE: folder_name is optional. For example, you can clone in alif_vscode-template-e1c instead of the default alif_vscode-template.

4. Open the alif_vscode-template project cloned above in your VSCode installation by running “Open Folder” from the File menu. Run the “First time pack installation” VSCode task.

F1 -> Tasks: Run Task -> First time pack installation

This will load the latest CMSIS DFP 2.0 pack and associated Arm CMSIS packs.

5. You need to set two variables in the user preferences file to point to the Segger GDB server and the Alif Security Toolkit.

F1 -> Preferences: Open User Settings (JSON)

Add the two lines highlighted below into the file and press CTL-S to save it. If you are not using the default directories for the Segger tools or the Alif Security Toolkit, edit those entries accordingly.

```
{
  "cmake.additionalCompilerSearchDirs": [
    "C:/msys64/mingw32/bin",
    "C:/msys64/mingw64/bin",
    "C:/msys64/clang32/bin",
    "C:/msys64/clang64/bin",
    "C:/msys64/clangarm64/bin",
    "C:/msys64/ucrt64/bin"
```

```

}
"cortex-debug.JLinkGDBServerPath": "C:/Program Files/SEGGER/JLink/JLinkGDBServerCL.exe",
"alif.setools.root": "C:/app-release-exec"
}

```

6. Make sure E1C is selected as the Active Target in the CMSIS Manage Solution. The M55 subsystem in the B1 chip is the same as the Ensemble E1C chip, so this selection is used for M55 code development in VS Code for either device. Select blinky as the Active Project in the CMSIS Manage Solution and click on the Build (hammer) icon to build the solution.
7. To program the blinky application on the DK-E1C board, run Program with Security Toolkit (select COM port) VSCode Task.
F1 -> Tasks:Run Task -> Program with Security Toolkit (select COM port)

Select the appropriate COM port (this needs to be done only once. Subsequent times, you can run *Program with Security Toolkit* VSCode task instead.
8. Go to the Segger website and download the latest version of the Segger tools. Debugging designs on the E1C and B1 DevKits has been tested using SEGGER J-Link GDB server V8.54, and the VS Code setting.json file entry for "cortex-debug.JLinkGDBServerPath" points to this new version.

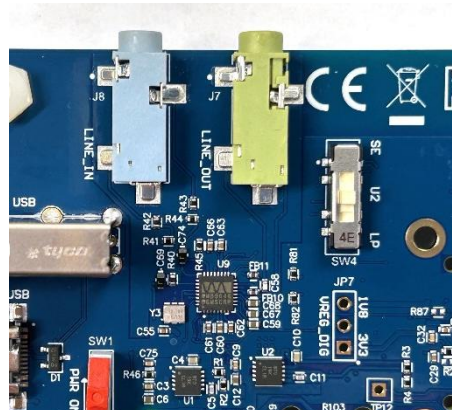
If you need help getting your design environment configured, file a technical support ticket at <https://alifsemi.com/support/technical-support/>

Alternate UART Port Selection

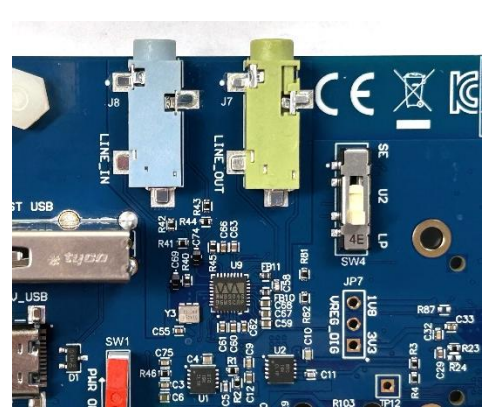
With the default switch setting for SW4, the COM port connection will be to the SEUART. If it is desired to connect to a user UART, the other switch positions can select UART2 or the LPUART. The images below show the alternate UART selections.



SEUART



UART2



LPUART

Configuration Jumpers

All Configuration Jumpers are male headers, 2.54mm pin spacing.

Note: As the kit is delivered, a jumper is installed at jumper post position JP9 and you can use the position as required. However, if any jumper post positions are required other than JP9, you must first cut the shunting trace on the PCB with a small blade that connects the jumper posts together and then solder posts onto the PCB. After this action, you can install and remove a jumper across the posts as required.

For the default Jumper settings, look for  RED Jumper Shunts. Factory default settings are denoted below with **.

1. JP1 – VDD_USB_3V3 Current Measurement
2. JP2 – VDD_BUCK Current Measurement
3. JP3 – SOC Voltage selection for VDD_MAIN and VDD_BUCK
 - a. Pin 1-2 SOC power = 3.3V
 - b. Pin 2-3 FLEXIO = 1.8V **
4. JP4 – VDD_MAIN Current Measurement
5. JP5 – VDDIO Current Measurement
6. JP6 – FLEXIO Voltage Selection
 - a. Pin 1-2 FLEXIO = 3.3V
 - b. Pin 2-3 FLEXIO = 1.8V **
7. JP7 – VREG_DIG_1V8 Selection
 - a. Pin 1-2 VREG_DIG_1V8 = 1 μ F
 - b. Pin 2-3 VREG_DIG_1V8 = 100nF **
8. JP9 – Click Board Voltage selection
 - a. Pins 1-2 Click Board I/O = 1.8V
 - b. Pins 3-4 Click Board I/O = 3.3V **
 - c. Pins 5-6 Click Board I/O = 5.0V